

A Security Analysis of an Activation-Steering LLM Watermark: Key Recovery, Forgery, Scrubbing and Keyed Defences

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Abstract. Activation-steering watermarks add a secret vector to a hidden layer during generation and detect it from the model’s internal activations when the text is re-read; their authors motivate the design by robustness to paraphrase. We present, to our knowledge, the first security analysis of such a watermark: the construction of LLM Self-Recognition ([Ardoyn, Schäfer, and Wunder 2026](#)), re-implemented on Qwen2.5-1.5B at 256 tokens and measured under 6 pre-registered protocols against a passive attacker who holds the open weights and up to 1,024 watermarked outputs but cannot query the detector. First, under our measures the published strength did not reproduce on 1–3B base models: no strength made the scheme’s trained probe reliably detectable without a perplexity cost. Second, that probe, re-trained by us, accepts steering that is not the key, so an attacker without the key produces fluent forgeries it accepts at the one strength where quality is acceptable (19.5% against a bar of 14.3%). Third, an exact key-resampling score test, a transfer of GaussMark’s statistic to an additive activation vector, attributes 96.0–100.0% of genuine texts at an exact 1% false-positive rate (exact on average over keys; every key’s rate checked) and rejects those forgeries. Fourth, the same statistic, averaged over 64 observed outputs, recovers the fixed key at the known layer (median cosine 0.940 and 0.951) and finds the layer for most keys when it searches, and its forgeries are attributed as often as genuine text (88.0 and 85.5% against 86.5 and 90.0% at the known layer; 86.0 and 87.5% with the layer search from 256 texts); an activation-mean estimator recovers nothing from 1,024. Fifth, an off-the-shelf paraphrase scrubs the test’s evidence at the quality-acceptable strengths (64.0 and 53.0% success) and not above them. Sixth, keying the direction by the previous h tokens or rotating among 8 keys, transfers of context hashing and key rotation from token-level watermarking that the scheme’s authors name as future work, blocks this attacker at $n \leq 1,024$: the context-keyed arms also improve perplexity but lose most of the paraphrase robustness, and rotation, whose quality is unchanged, falls to a clustering attacker from 64 texts at $\rho = 0.50$ (and, post hoc, at 0.35). For the fixed key on this model, the strengths that are detectable and paraphrase-robust carry a quality cost (perplexity ratios 1.62 and 2.29), the quality-neutral strengths are scrubbed, and the published detector type is forgeable without the key at the quality-neutral strength (and, by plain acceptance, above it). The protocols, locks and code accompany the paper.

1 Introduction

Activation-steering watermarks mark text from inside the model: during generation a secret vector is added to one hidden layer at every token, and at detection the finished text is re-encoded with the same model and the signal is read from its activations rather than from token statistics. Their authors motivate the design by robustness to rewriting: for LLM Self-Recognition ([Ardoyn, Schäfer, and Wunder 2026](#)), the scheme analysed here, “Our method shows greater robustness to paraphrasing” (p. 5), and detection is white-box by construction (“we assume white-box access to the internal activations of M ”, p. 2).

Whether such a watermark is secure has not been measured. Its authors state the risk and leave it open: “if an attacker can recover the secret steering vector and target layer, the watermark could be trivially reproduce to spoof the identity of the steered LLM”, and they propose as future work a vector that “rotates or evolves during generation according to a pseudo-random schedule” (p. 7). Stealing, forgery and scrubbing of token-level watermarks have an established literature ([Jovanović, Staab, and Vechev 2024](#); [Pang et al. 2024](#); [Gu et al. 2024](#); [Zhang et al. 2026](#)), and a September 2026 evaluation of it excludes model-internal schemes by scope ([Li et al. 2026, 2](#)). To our knowledge, no prior work measures key recovery, forgery or scrubbing for an activation-steering watermark; search absence is not proof of novelty.

Regulators now name the properties measured here: the European Commission’s Code of Practice on marking and labelling of AI-generated content (final, 10 June 2026) ([European Commission 2026](#)) expects an imperceptible watermark on free-form text longer than 200 tokens (Sub-measure 1.1.2, p. 9), measures *reliability* by “the error rate of the detection” (Measure 3.2, p. 17) and asks that solutions “maintain intended performance levels under varying conditions, covering both common alterations and adversarial attacks”, paraphrasing among them (Measure 3.3, pp. 17–18), and NIST AI 100-4 ([National Institute of Standards and Technology 2024](#)) defines a *secure* watermark as one that resists attempts “to remove or tamper with the watermark information, or to insert forged watermarks” (p. 13). A false-positive rate (FPR) at a fixed level, detection after paraphrase and resistance to forged watermarks are the three measurements below (the quoted measures are in Appendix I); we use these texts for vocabulary and motivation only, and nothing here is a statement about any provider’s compliance or deployment.

What we do. We re-implement Self-Recognition’s construction on Qwen2.5-1.5B at 256 tokens (Section 2, Section 4) and measure it against a passive attacker who holds the open weights and up to $n = 1,024$ watermarked outputs, text only, and cannot query the detector (Section 2.2), under 6 pre-registered protocols locked by hash before any outcome existed (Section 4, App. A). Because the published strength does not transfer across models, strength is calibrated first as ρ , the vector’s norm relative to the activation norm, along the detectability–quality curve (Section 5.1). Five questions follow, one per results subsection: forgery of the published probe without the key (Section 5.2), the exact key test (Section 3, Section 5.3), key recovery from outputs (Section 5.4), scrubbing at matched quality (Section 5.5) and keyed directions with their price (Section 5.6).

Contributions. Every claim is bounded to one model (Qwen2.5-1.5B base), one layer, 8 keys, 256-token continuations, the strengths tested and this attacker family with $n \leq 1,024$; rates are medians over keys with cluster-bootstrap intervals and per-key tables (App. D).

1. To our knowledge, the first security analysis of an activation-steering LLM watermark (key recovery, forgery, scrubbing, an exact test and a keyed defence), on one published scheme, re-implemented; SLAM and AWM are discussed, not tested.
2. The scheme’s published detector type, a trained per-token probe re-implemented and re-trained by us, accepts steering that is not the key (median cosine at most 0.062 with it) and degenerate text; fluent, non-repetitive forgery without the key is practical at $\rho = 0.35$ (fluent acceptance 19.5% against a bar of 14.3%) and inconclusive at 0.50 and 0.70. The authors note that paraphrased human text spoofs their detector (p. 5), and under our conditions their published setting did not reproduce on 1–3B base models (Section 5.1, with their numbers beside ours).
3. An exact key-resampling score test, a transfer of GaussMark’s statistic ([Block, Rakhlin, and Sekhari 2025](#)) to an additive activation vector, attributes 96.0–100.0% of genuine texts at an exact 1% false-positive rate (exact on average over keys; every study key calibrated), beats the probe at $\rho \leq 0.50$, attributes random-key text at a median rate of 0.0% (single keys up to 4%), and makes those forgeries not practical for the median key.
4. The first measured key-recovery attack on such a watermark (search level): the test’s own statistic, averaged over 64 outputs, recovers the fixed key at the known layer (median cosine 0.940 and 0.951 at $\rho = 0.35$ and 0.50; the layer search finds it for 6 and 6 of 8 keys), and its forgeries are attributed as often as genuine text (88.0 and 85.5% against 86.5 and 90.0% at the known layer; 86.0 and 87.5% with the layer search from 256 texts), where the activation-mean estimator recovers nothing from 1,024 (cosine at most 0.16). That the detector’s best statistic is the thief’s best estimator is a measured fact with an inferential explanation (Section 3); the token-level budget comparison carries an access caveat (Section 5.4).

5. To our knowledge the first scrubbing evaluation of such a watermark under an exact, per-key-calibrated test at matched quality (the authors’ own paraphrase figures are robustness measurements, not attack evaluations; Section 6), with an insider (true-key) bound: an off-the-shelf paraphrase succeeds 64.0 and 53.0% of the time at $\rho = 0.25$ and 0.35 and is not effective at 0.50 and 0.70; the comparison with the authors’ robustness figures is indirect.
6. The measured trade-off along ρ for the fixed key on this model (an interpretation, labelled): the detectable and paraphrase-robust strengths cost perplexity (ratios 1.62 and 2.29), the quality-neutral strengths are scrubbed, and the probe is forgeable at the one strength where quality is acceptable; we are not the first to note a trade-off (Section 6).
7. Context keying ($h = 1, 4$) and rotation among 8 keys, transfers of context hashing ([Kirchenbauer et al. 2023, 2024](#)) and key rotation ([Kirchenbauer et al. 2023](#); [Pang et al. 2024](#); [Gu et al. 2024](#); [Dathathri et al. 2024](#)) to activation steering, named as future work by Ardoin et al. (p. 7): both hashed arms block this attacker at $n \leq 1,024$ (forgeries attributed at most 9.0%) and improve perplexity, but lose most of the paraphrase robustness (material for $h = 1$ at both strengths and $h = 4$ at 0.50, inconclusive for $h = 4$ at 0.35); rotation keeps robustness and falls to a clustering attacker from 64 texts at $\rho = 0.50$ (95% attributed), and at 0.35 only through a labelled post-hoc addendum beside the pre-registered “blocked” verdict (Section 5.6).

Two practical points are restated for this family, as others have noted: a deployer with one key should check its own false-positive rate on paraphrased null text, since resampling p-values are exact only on average over keys (Section 5.5), and a quality bar for forgery or scrubbing must pair perplexity with a repetition term and be read against reference samples (Section 2.2). Per-key heterogeneity, the probe’s acceptance of paraphrases at high strength and the reading that the probe detects steering rather than the key are a descriptive, a lead and an inference, labelled where they appear. We do not claim the first stealing attack on LLM watermarks, a new watermark, immunity to stealing, a fix of the trade-off, or results beyond the model, budgets and attackers tested (Section 7). The protocols, locks and code accompany the paper (Section 8).

2 Background and threat model

2.1 The scheme

An activation-steering watermark adds a secret vector to the model’s hidden state during generation and detects it by re-encoding the finished text with the same model and reading the hidden state back. We analyse the construction of Self-Recognition ([Ardoin, Schäfer, and Wunder 2026](#)). Given a direction $v \in \mathbb{R}^d$ and a coefficient $\alpha > 0$, its authors “add the scaled vector to the intermediate activation at layer l at each token generation step” (p. 2), $A_l(x_i) \leftarrow A_l(x_i) + \alpha v$. The direction is random and sparse, “sampled from $U([-1, 1]^d)$, with 99.7% of dimensions set to zero” (p. 11), so a key on a model of width $d = 1536$ has $k = 4$ non-zero coordinates; the published strength is $\alpha = 5$, and “Injection and detection are performed at the middle layer” (p. 11). Detection is white-box (“we assume white-box access to the internal activations of M ”, p. 2): the text is re-encoded and a multilayer perceptron (MLP) probe on the layer- l activations decides, “The MLP has two hidden layers of width 32, trained for a single epoch to predict the steering index k of a given extracted activation” (p. 3), with token decisions aggregated to a text decision; a training-free alternative is the cosine similarity between the activations and the key (p. 6).

We re-implement the scheme on Qwen/Qwen2.5-1.5B (Section 4) and the detector as the authors’ public code defines it: an MLP on raw per-token activations at the steered layer, trained for one epoch on the owner’s own watermarked and unwatermarked texts and scored as the mean token probability of the watermarked class (App. A). An owner must fix a false-positive rate, so the score is thresholded at the 99th

percentile of the owner’s own unwatermarked texts (nominal 1%), and we report acceptance at that threshold rather than the accuracies the authors report. Two definitions are ours. **Strength:** α does not transfer across models, because activation norms differ by model and layer, so we measure $\rho = \|v\|/N$, with N the median token activation norm at the steered layer over unsteered text ($N = 56.90$ here); the published $\alpha = 5$ maps to very different ρ on the models we tried (Section 5.1). **Working range:** at the strengths studied, $\rho \in \{0.25, 0.35, 0.50, 0.70\}$, the probe does not detect the lower strengths reliably and the higher strengths cost perplexity (Section 5.1), so the attacks run along that curve rather than at one setting.

2.2 Threat model

We follow Kerckhoffs’s principle ([Kerckhoffs 1883](#)), as the nearest keyed activation scheme does ([Aremu et al. 2026](#)): the weights, the scheme (including k and the candidate layers), ρ and the detector’s form are public; the direction v , the layer l and the owner’s private detector data (the trained probe and its threshold; for the exact test of Section 3, nothing beyond v and l) are secret.

The **attacker** is passive. It holds the open weights and observes n watermarked continuations at one strength, text only, without their prompts ($n \in \{64, 256, 1024\}$ for forgery, $n \in \{1, 4, 16, 64, 256, 1024\}$ for recovery). It can generate unlimited unwatermarked text on its own prompts and re-encode any text through the open model, so it sees activations and gradients at every layer. It cannot query the owner’s detector and never sees v , the probe, its threshold, the owner’s reference texts or the evaluation prompts; the layer l is withheld too, except in two readings that say so, the fixed key’s known-layer estimate (Section 5.4) and the keyed arms (Section 5.6); the attack code receives features and public parameters only, which every protocol’s validation asserts (App. A). The **owner** knows v and l , calibrates its detector on its own unwatermarked texts and sees only the text at detection time.

Three goals are measured, in the terms of the token-level security literature ([Li et al. 2026, 3](#)): stealing “aims to recover reusable watermark-related information”; “Scrubbing aims to transform a watermarked text y_w into an attacked text y_a that is no longer detected as watermarked while preserving text quality”; “Spoofing aims to generate a forged text y_s that is accepted as watermarked without being produced by the genuine watermarking process”.

- **Recovery:** $\cos(\widehat{v}, v)$ between the estimate and the key, whether the estimated layer equals l , and the share of the support recovered.
- **Forgery** (spoofing): new text, generated by the attacker on prompts it does not control, counts only if the owner’s detector **accepts** it at the 1% threshold **and** it is as fluent and as non-repetitive as the owner’s genuine watermarked texts (perplexity and seq-rep-4 ([Welleck et al. 2020](#)) at most the genuine 95th percentiles; Section 4); the share meeting all three conditions is the **fluent acceptance** (FA). Forgery is **practical** at a strength if, at some n , the median FA over keys reaches half the genuine FA and its lower confidence bound exceeds the upper bound of a **random-key control** (a fresh key of the same distribution and norm: what generic steering alone achieves). This is the quality-constrained success of the token-level evaluations ([Li et al. 2026](#)), with proxies in place of a judge.
- **Scrubbing:** a watermarked text is transformed; the transformation **succeeds** if the detector no longer detects it **and** four quality conditions hold (fluency, repetition, length, meaning; Section 4), against the detector’s miss rate on the untouched originals or random edits of the same size.

The budgets differ in kind from the token-level literature, whose attackers query a black-box API with “ $n = 30,000$ responses of token length ≤ 800 ” ([Jovanović, Staab, and Vechev 2024, 6](#)), “2.2 million tokens in total” ([Pang et al. 2024, 6](#)) or “2,000 victim watermarked texts” ([Zhang et al. 2026, 8](#)) and “has

only blackbox access to full generations” (Jovanović, Staab, and Vechev 2024, 3); ours has at most 1,024 texts of 256 tokens but re-encodes them through the model that produced them. A perplexity bar alone admits degenerate text, since “a highly repetitive, degenerate loop often yields low PPL” (Ardoin, Schäfer, and Wunder 2026, 14), and SLAM scores quality on four axes (Harel-Canada and Sahai 2026); our first fluency rule was perplexity only and most of its accepted forgeries at the strongest setting were loops (an exploratory reading, labelled; Section 5.2, App. E), so every later rule pairs perplexity with seq-rep-4 on both sides, the attacker’s self-check and the evaluation, and fixed samples of passing texts are read (App. H). Both points have been made before; we restate them for this family.

2.3 Attack routes

Three routes are compared against the attacker’s own unwatermarked reference texts (pool C, 2,000 texts; Section 4), coordinate by coordinate and layer by layer, as z-scores of the mean difference of a per-text feature (App. C).

Route A (key recovery from activations) takes the layer with the largest $|z|$ of the token-mean activation, the k coordinates with the largest $|z|$ there as the support and the mean differences as values, rescaled to ρ times the attacker’s own median norm.

Route B (footprint imitation) estimates no key and copies the dense mean difference at the layer with the largest z-norm, rescaled.

Route A’ (key recovery from the detector’s statistic) applies Route A’s rule to the standardised gradient of the continuation log-likelihood with respect to an additive vector at each layer, the exact test’s own statistic (Section 3), averaged over the observed texts, per observable context against context-keyed directions and per cluster (or on the whole mixture) against rotation.

From Study 1 v0.3 the attacker screens five candidate layers with 16 trial generations each, keeping those no less fluent than the observed texts (and, from v0.4, no more repetitive), since it cannot query the detector.

The **scrubbing attackers** are two off-the-shelf paraphrasers (P-Qwen, P-Phi; Section 4) with an attacker-side quality self-check and up to three attempts, and key-guided word substitutions at 2, 5 and 10% of the tokens steered by the true key (E-true, an insider bound), by the Route A estimate (E-est) or at random (E-rand).

Forgeries are 100 continuations per condition on fixed evaluation prompts (pool D), scored by the owner’s detector beside the **oracle** (the true key at the true layer) and **random-key** controls on the same prompts. A rule asks at every strength whether the random key alone reaches half the oracle’s rate (“generic steering suffices”), in which case forgery would need no key.

3 An exact key test for additive activation vectors

The probe of Section 2.1 detects a learned footprint, on our reading of Section 5.2. A detector of the *key* needs no training, as a transfer of GaussMark’s score test (Block, Rakhlin, and Sekhari 2025). GaussMark watermarks by perturbing weights, “Sample watermark key $\xi \sim N(0, \sigma^2 I)$ ” and “Watermark model by setting $\theta(\xi) = \theta + \xi$ ” (Alg. 1, p. 6), “in only a single feedforward layer” (p. 5), and detects with $\psi(y, \xi | x) = \langle \xi, \nabla_{\theta} \log p_{\theta}(y | x) \rangle / (\sigma \| \nabla_{\theta} \log p_{\theta}(y | x) \|)$, rejecting when $\psi \geq \Phi^{-1}(1 - \alpha)$ (Alg. 2, p. 7), because “the distribution of $\psi(y, \xi | x) \dots$ is always a standard normal under H_0 because y is independent of ξ ” (p. 9). This is the score statistic for the perturbation (Rao 1948): to first order, text generated under the perturbed model shifts the expected gradient along the perturbation. In an activation-steering watermark the perturbation is an additive vector b at the output of layer l , so the statistic is the gradi-

ent of the text’s log-likelihood with respect to b at $b = 0$, projected on the key. Two things change the calibration: the keys are sparse with uniform values, not Gaussian, so the null law cannot be read from Φ ; and the gradient’s coordinates have very different scales, so a raw projection depends on a few outlier dimensions. We therefore standardise and resample keys.

The statistic S4 and its exact p-value. One forward and one backward pass over a text’s first 256 continuation tokens (no prompt) gives $g = \partial \log p(y) / \partial b|_{b=0} \in \mathbb{R}^{1536}$ at layer 14. The owner centres and scales each coordinate against its own unwatermarked reference texts (pool F, 200 texts), $x = (g - \mu_F) / \sigma_F$, and computes $T(x, v) = \langle x, v \rangle / (\|x\| \|v\|)$. With $M = 999$ null keys $v_1, \dots, v_M$ drawn from the scheme’s public key distribution (fixed seeds, disjoint from every other key), $p = \left(1 + \#\{j: T(x, v_j) \geq T(x, v)\}\right) / (1 + M)$ (Phipson and Smyth 2010), and a text is attributed to v when $p \leq 0.01$. The true-positive rate (TPR) is the share of genuine watermarked texts attributed to their key. If the text was produced independently of v (unwatermarked, human, or steered by another key), v and the null keys are exchangeable, so $P(p \leq \alpha) \leq \alpha$ **exactly, on average over the draw of the key**. GaussMark’s authors state the same limit of their guarantee: the bounds “are valid for any fixed $x \in \mathcal{X}$, but are marginalized over the key ξ ” (p. 11). A deployer holds one key, whose own false-positive rate need not equal α ; we never claim exactness for every key. The protocols therefore add a **per-key calibration gate**: a key is calibrated if its false-positive rate on 1,000 of the owner’s unwatermarked texts is at most 3%, and the pooled rate must lie in [0.3, 2.5]%. On the study keys the pooled rate was 0.81%, 8 of 8 keys were calibrated (0.1–1.7%), and the pooled rate on human continuations was 0.68% (Table A3).

Why standardise. A pilot on four tuning keys compared, under the same resampling test, the raw cosine of the token-averaged activations (S1, the scheme’s training-free statistic), its standardised form (S2), the raw gradient score (S3) and the standardised gradient score (S4): the gradient statistics detected 96–98% (S3) and 94–98% (S4) of single texts at an exact 1% false-positive rate at the working strengths, the raw cosine 0% (Table A2). It also exposed the per-key issue: because the raw activation mean is nearly the same vector for every text, a key’s rank among the null keys is almost fixed, and over 1,000 random keys S1 fired on more than half of the null texts for 1.1% of keys while its mean rate stayed 1.15% (an exploratory diagnosis, App. B); standardisation kept every key at most 4.75% (S4). S4 was chosen as the primary statistic by decision after that diagnosis; the pilot’s pre-set selection rule is reported as written in App. B.

Two variants used by the defence (Section 5.6). For **rotation** among K keys the owner tests the union, $U = \max_k \langle x, v_k \rangle$, with the null from 999 disjoint groups of K public null keys, so the level is exact by the same argument: a transfer of the multi-key design of Kirchenbauer et al. (Kirchenbauer et al. 2023), “we run k tests, one for each of the possible keys” (p. 7), with a resampled maximum in place of a Bonferroni correction. For a **context-keyed** direction $v(c_t)$, derived by a keyed pseudorandom function from the secret and the previous h tokens, the owner takes the per-position gradients g_t (one backward pass), standardises them with pooled position statistics from pool F and sums the projections on each position’s own key, $S(x, m) = \sum_t \langle x_t, v_m(c_t) \rangle$, with the p-value from 999 resampled secrets; the contexts are read from the text itself (App. B).

Why the same statistic is the thief’s estimator (an inference; the measurement is Section 5.4). To first order the expected gradient of a text steered by v is $\mathbb{E}[g] \approx Fv$, with F the Fisher information of b at 0, which is what gives $\langle g, v \rangle$ its power; the same relation says that the **mean** of g over many observed outputs estimates Fv , and after the per-coordinate standardisation the attacker performs against its own reference texts, a sparse estimate of that mean points along v . Route A’ (Section 2.3) is this estimator; it needs the open weights and the observed texts, not the detector. App. B sketches the first-order argument and its assumptions; we present it as the reason we expected the recovery, not as a proof. On the tuning keys the

estimator reached a median cosine of 0.80 from 4 texts and 0.95 from 16 at $\rho = 0.35$, 0.93 from 64 at $\rho = 0.50$ and 0.57 from 100 at $\rho = 0.70$, where Route A’s activation mean reached at most 0.087 (Table A5); the study keys follow in Section 5.4.

4 Experimental setup

Model and keys. The target is Qwen/Qwen2.5-1.5B (base, revision `8faed761`, apache-2.0) (Qwen Team 2024a, 2024b) in bfloat16 on Apple silicon; $d = 1536$, 28 decoder layers; steering and detection at the output of layer 14, the middle layer. Keys follow the scheme’s construction, $k = 4$ non-zero coordinates ($[0.003 \cdot d]$) with values uniform in $[-1, 1]$, rescaled to norm ρN , $N = 56.90$. The **study keys** are the seeds 1001–1008 (8 keys), first used in Study 1 v0.2 and reused unchanged by every later study; the **random-control keys** are 5001–5008; four **tuning keys** (9001–9004) served the calibrations and pilots and appear in no study result. Strengths are $\rho \in \{0.25, 0.35, 0.50, 0.70\}$ (Study 5: 0.35 and 0.50). Generation uses the scheme’s sampling (temperature 0.7, top-p 0.9, top-k 50, repetition penalty 1.1) and exactly 256 new tokens; every detector and attacker feature is computed from the continuation only.

Prompts and pools. Prompts are the first 40 words of documents from the C4 realnewslake validation shard (Raffel et al. 2020) (13,863 records, 0 parse failures, 8,580 documents of at least 240 words), with the next 200 words kept as the **human continuation**. A fifth of the units, by ID hash, is a **holdout** (1,684 units) sealed in the loader, which refuses to serve it without a final protocol lock; it was never used. The 6,896 development units are assigned once, by hash rank, to disjoint pools A–F, whose sizes and roles are in Table 1’s note: the owner’s unwatermarked null texts with their human continuations (A), the attacker’s observed and reference prompts (B, C), the evaluation prompts for forgeries and controls (D) and the owner’s probe-training texts (E, F). The owner’s training texts are disjoint from its threshold, check and evaluation texts, and the attacker’s prompts from all of them. WikiText-103 (Merity et al. 2017) was frozen as a second domain and is not used by the results reported.

Detectors. The probe is trained per key and strength on the layer-14 token activations of pool E against pool F, scored as the mean token probability and thresholded at the 99th percentile of pool A1 (nominal 1% false-positive rate, checked on A2). The exact test S4 (Section 3) uses $M = 999$ null keys at $p \leq 0.01$, with a stricter secondary reading at 9,999 null keys and $p \leq 0.001$. Both see the continuation only.

Paraphrasers, editors and quality proxies. P-Qwen is Qwen2.5-1.5B-Instruct (Qwen Team 2024c) and P-Phi is Phi-3.5-mini-instruct (Abdin et al. 2024), each given the continuation with the fixed paraphrase instruction `PROMPTS["P2"]` from `research/study2_pilot/scrub_core.py`, selected by `research/study2/common_s2.py`, in its chat template (App. C), sampled at temperature 0.7 and top-p 0.9 with at most 400 new tokens; the attacker keeps the first of up to three attempts that passes its own quality check. The editor replaces whole words by likely alternatives at the positions a first-order estimate marks as contributing most to the detector’s statistic, in rounds of 5% of the tokens, saved at 2, 5 and 10% (App. C). Quality is measured by proxies, not by human judgement: perplexity under the unsteered model given the prompt; seq-rep-4 (Welleck et al. 2020); token length; and the cosine between all-mpnet-base-v2 sentence embeddings (Reimers and Gurevych 2019) of a scrubbed text and its original. A forgery is fluent if its perplexity and seq-rep-4 are at most the 95th percentiles of the owner’s genuine watermarked texts at that key and strength. A scrub passes if its perplexity stays within $\max(1.25 \times \text{the original’s}, \text{the 95th percentile of pool A’s human continuations})$ and its seq-rep-4 within $\max(\text{the original’s}, \text{that 95th percentile})$ (perplexity 25.4, seq-rep-4 0.048), its length is at least 80% of the original’s and its embedding cosine exceeds 0.825, the 95th percentile over pool A’s same-prompt pairs of a model continuation and the human continuation; the attacker’s self-checks use the same forms on its own pool C data (perplexity 22.2, seq-rep-4 0.048, cosine 0.810). Stricter readings (the model-text 95th percentile, perplexity 7.3; the

human median, 12.9) are secondaries, and fixed samples of forged, scrubbed and genuine texts are read (App. H).

Uncertainty. The key is the unit of independence; texts are nested in keys. Every rate is the median over the 8 keys of the per-key rate, with a 95% interval from a cluster bootstrap over keys and then texts ($B = 2,000$) (Davison and Hinkley 1997); contrasts between two detectors or two methods on the same texts use paired resampling. Beside every median we report the number of keys meeting the bar, with per-key tables in App. D, because a median can pass while a few keys fail completely; counts of keys carry Clopper–Pearson intervals (Clopper and Pearson 1934).

Pre-registration and compute. Each study has a protocol fixing its question, information timing, metrics, materiality thresholds, rules and stop rules, and code validated on inputs only (synthetic data and the tuning keys) before any outcome exists; the protocol, the code and every reused input file are hashed into a lock file, and the runners verify every hash before and after each phase and refuse otherwise. A correction becomes a new version with the earlier verdicts standing (Study 1 v0.2 $\rightarrow$ v0.3 $\rightarrow$ v0.4); every rule’s bounds were checked at the extremes before locking; readings added after a run are labelled exploratory or post hoc wherever they appear. **Table 1** lists the 6 locked studies with their rules, bars, design, lock dates and run times; App. A gives every pilot and specification (Table A1). Everything ran on one Apple-silicon laptop (M5, 16 GB unified memory) with PyTorch 2.14.0; the six runs took 91.1 h in total (22.4, 6.4, 6.5, 2.9, 26.6 and 26.3 h), with model weights, prompt corpora and paraphraser pinned by revision and file hash.

Table 1. The six locked studies: question, pre-registered rules and bars, design, lock and run

Study	Question	Pre-registered rules	Materiality bars	Keys; strengths ρ	Attacker budget n	Locked (UTC date); protocol SHA-256	Commit	Run (h)	Results
Study 1 v0.2	Key recovery and forgery against the trained probe, along p	G1 null calibration; G2 attack isolation; L1 working watermark; R1 recovery; R2 forgery practical; R3 generic steering suffices	R1: median $\cos \geq 0.9$ and layer hit for ≥ 7 of 8 keys; R2: acceptance $\geq 50\%$ of the oracle’s and the lower bound above the random-key control	1001–1008; $\rho \in \{0.25, 0.35, 0.50, 0.70\}$	recovery {1, 4, 16, 64, 256, 1,024}; forgery {64, 256, 1,024}	2026-09-26; 8794f4ddb25e	99d97978	22.4	research/outputs/study1_v0.2/results.json
Study 1 v0.3	Fluent forgery (perplexity condition) with the attacker’s own layer search	integrity check; F1 fluent forgery practical; F3 fluent generic steering suffices	F1: fluent acceptance $\geq 50\%$ of the oracle’s and the lower bound above the random-key control	1001–1008; $\rho \in \{0.35, 0.50, 0.70\}$	{64, 256}	2026-09-27; 842b71fc4624	8b8718de	6.4	research/outputs/study1_v0.3/results.json
Study 1 v0.4	Fluent, non-repetitive forgery (perplexity and seq-rep-4 conditions on both sides)	integrity check; F1; F3	F1 as v0.3 with the repetition term	1001–1008; $\rho \in \{0.35, 0.50, 0.70\}$	{64, 256}	2026-09-28; ab240e341a6c	6c6236be	6.5	research/outputs/study1_v0.4/results.json
Study 3	The exact key test: power, per-key calibration, Study 1’s forgeries re-scored	G1 pooled FPR; G2 per-key calibration; P1 working test; X1 forgery practical; X3 generic steering; E1 power against the probe; H1 human text	G1: pooled FPR in [0.3, 2.5]%; G2: per-key FPR $\leq 3\%$; P1: TPR $\geq 20\%$; X1: exact-FA $\geq 50\%$ of the oracle’s	1001–1008; $\rho \in \{0.25, 0.35, 0.50, 0.70\}$	Study 1’s texts re-scored	2026-09-30; 26025c182d56	323c7a70	2.9	research/outputs/study3_v0.1/results.json
Study 2	Scrubbing under the exact test at matched quality:	G1, G2 per paraphraser; P1; S1 paraphrase effective; S2 edits	S1, S2: success $\geq 50\%$ and the lower bound above the comparator (the	1001–1008; $\rho \in \{0.25,$	Study 1’s texts scrubbed	2026-10-01; eba198a7ec4a	070075dd	26.6	research/outputs/study2_v0.

Study	Question	Pre-registered rules	Materiality bars	Keys; strengths ρ	Attacker budget	Locked (UTC date); protocol SHA-256	Commit	Run (h)	Results
	paraphrase and key-guided edits	effective; C1 the estimate helps	originals' miss rate; random edits)	0.35, 0.50, 0.70}					1/results.json
Study 5	Context-keyed (h = 1, 4) and rotating (K = 8) directions against the gradient-averaging attacker	G1, G2; P0 the threat; P1 usability; D1 stealing; D2 robustness cost; D3 quality cost	D1: FA $\geq$ 50% of the arm's genuine FA; D2: 20 points of scrub success; D3: perplexity ratio 1.10	1001–1008; $\rho \in \{0.35, 0.50\}$	{64, 256, 1,024}	2026-10-02; 3695f434a1b0	cf6c1b4b	26.3	research/outputs/study5_v0.1/results.json

Every hash shown equals the study's `PRE_RUN_LOCK.json`; the protocol files, the locked code files and the results files were re-hashed when this table was built and equal the locked values. Pilots and specifications are in Table A1.

Prompt pools (C4 development units assigned once by hash rank; §4): A 1,000 (the owner's unwatermarked null texts and their human continuations; A1 for the probe's threshold, A2 for its false-positive check), B 1,024 (the attacker's observed texts), C 2,000 (the attacker's unwatermarked reference texts), D 100 (the evaluation prompts), E 200 and F 200 (the probe's training texts; the per-position statistics).

5 Results

Each subsection reports one study's pre-registered rules as applied, in the study's own vocabulary (practical, effective, blocked, material, inconclusive), with exploratory and post-hoc readings marked; every number is generated from the locked outputs (App. A).

5.1 Calibration: no detectable-and-fluent strength for the probe on 1–3B base models

The published setting was re-run with the authors' probe on three base models and the tuning keys (Table 2; Figure 1). Its $\alpha = 5$ is a different strength on every model: $\rho = 0.10$ on Qwen2.5-1.5B, where no detector exceeded an area under the receiver operating characteristic curve (AUROC) of 0.57 and perplexity was unchanged; $\rho = 1.29$ on Llama-3.2-1B, where the probe reached AUROC 1.00 while the median perplexity rose from 5.6 to 32.6 and the sampled texts read as damaged (App. H); and $\rho \approx 0.67$ on Llama-3.2-3B. With strength measured as ρ , the three models trace one curve: the probe passes AUROC 0.95 from $\rho = 0.50$ on Qwen2.5-1.5B and 0.70 on Llama-3.2-1B, and never on Llama-3.2-3B (0.86 at $\rho = 0.70$), while the perplexity ratio stays within 1.25 only up to $\rho = 0.35$ on all three (512-token texts change neither verdict for the two models that have them; Table 2). The pre-set rule, the smallest ρ with AUROC ≥ 0.95 and a ratio ≤ 1.25 , returned no operating point in any of the 5 calibration runs.

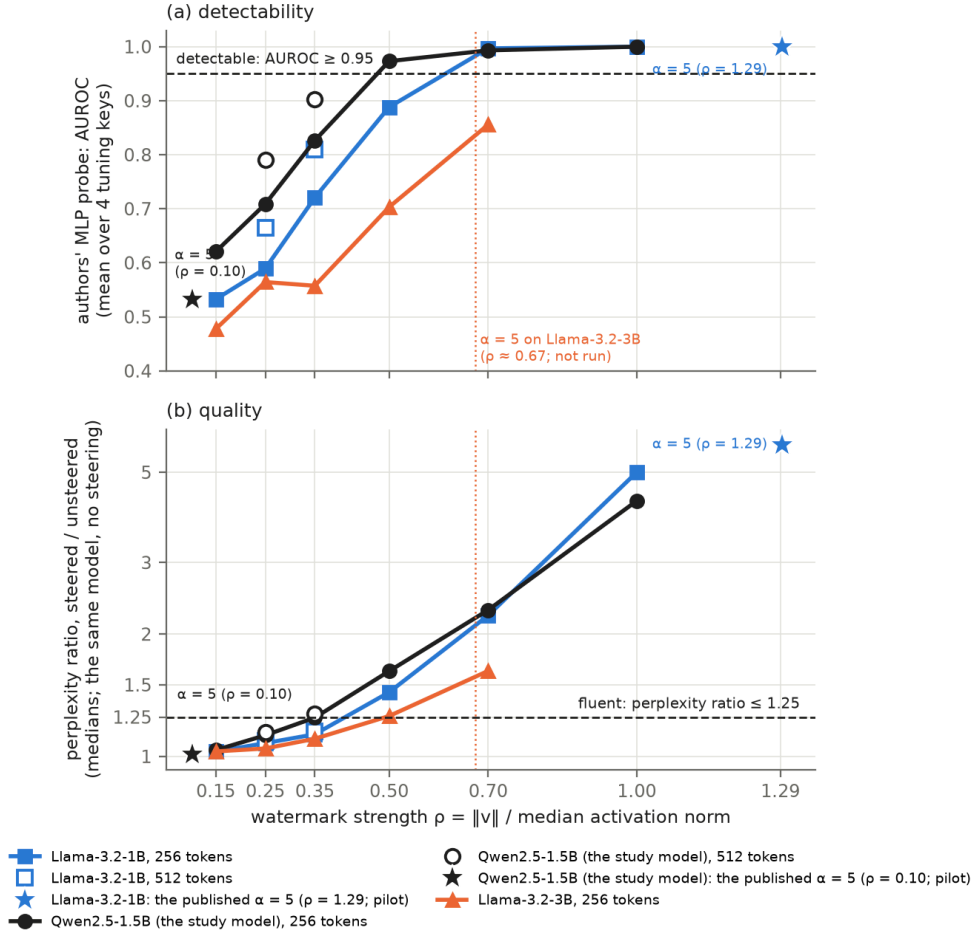


Figure 1. Calibration on three base models at 256 tokens: (a) the trained probe’s AUROC and (b) the perplexity ratio to unsteered text against the strength ρ , with the 0.95 and 1.25 bars and the published $\alpha = 5$ marked as ρ for each model; 512-token points for two models. No model crosses both bars at the same ρ .

Table 2. Calibration of the published scheme on three base models with strength measured as ρ (tuning keys 9001–9004; 40 test texts per key; the reproduction pilot’s $\alpha = 5$ rows as ρ): no strength is both probe-detectable and perplexity-neutral

model (base)	tokens	$\rho = \ v\ / N$	α equivalent	authors' MLP probe AUROC (mean over 4 tuning keys)	perplexity ratio (steered / unsteered, medians)	detectable (AUROC ≥ 0.95)	fluent (ratio ≤ 1.25)
Llama-3.2-1B	256	0.15	0.6	0.53	1.03	no	yes
Llama-3.2-1B	256	0.25	1.0	0.59	1.08	no	yes
Llama-3.2-1B	256	0.35	1.4	0.72	1.13	no	yes
Llama-3.2-1B	256	0.50	1.9	0.89	1.44	no	no
Llama-3.2-1B	256	0.70	2.7	1.00	2.22	yes	no
Llama-3.2-1B	256	1.00	3.9	1.00	5.00	yes	no
Llama-3.2-1B	512	0.25	1.0	0.66	1.09	no	yes
Llama-3.2-1B	512	0.35	1.4	0.81	1.15	no	yes
Llama-3.2-1B	256	published $\alpha = 5 \rightarrow \rho = 1.29$	5.0	1.00 (pilot; best detector 1.00)	5.85 (median perplexity 32.6 vs 5.6)	yes	no
Qwen2.5-1.5B	256	0.15	7.4	0.62	1.04	no	yes
Qwen2.5-1.5B	256	0.25	12.3	0.71	1.13	no	yes
Qwen2.5-1.5B	256	0.35	17.2	0.83	1.25	no	yes

model (base)	tokens	$\rho = \text{lvl} / N$	α equivalent	authors' MLP probe AUROC (mean over 4 tuning keys)	perplexity ratio (steered / unsteered, medians)	detectable (AUROC ≥ 0.95)	fluent (ratio ≤ 1.25)
Qwen2.5-1.5B	256	0.50	24.5	0.97	1.62	yes	no
Qwen2.5-1.5B	256	0.70	34.3	0.99	2.29	yes	no
Qwen2.5-1.5B	256	1.00	49.0	1.00	4.25	yes	no
Qwen2.5-1.5B	512	0.25	12.3	0.79	1.15	no	yes
Qwen2.5-1.5B	512	0.35	17.2	0.90	1.27	no	no
Qwen2.5-1.5B	256	published $\alpha = 5 \rightarrow \rho = 0.10$	5.0	0.53 (pilot; best detector 0.57)	1.01 (median perplexity 5.5 vs 5.5)	no	yes
Llama-3.2-3B	256	0.15	1.1	0.48	1.03	no	yes
Llama-3.2-3B	256	0.25	1.9	0.56	1.05	no	yes
Llama-3.2-3B	256	0.35	2.6	0.56	1.11	no	yes
Llama-3.2-3B	256	0.50	3.7	0.70	1.26	no	no
Llama-3.2-3B	256	0.70	5.2	0.86	1.62	no	no
Llama-3.2-3B	256	published $\alpha = 5 \rightarrow \rho \approx 0.67$ (not run; nearest grid point 0.70)	5.0	—	—	—	—

N is the median token activation norm at the steered layer over unsteered text (Llama-3.2-1B: layer 8 of 16, $N = 5.67$; Qwen2.5-1.5B: layer 14 of 28, $N = 56.90$; Llama-3.2-3B: layer 14 of 28, $N = 12.67$). The operating-point rule (the smallest ρ with AUROC ≥ 0.95 and ratio ≤ 1.25) returned none on every model at either length. Perplexity is the continuation's under the same model without steering, given the prompt; the pilot's perplexity column was added after that pilot's rule was fixed (exploratory).

This is a result under our conditions, not a test of the authors' claim. They report "no quality degradation" (p. 1) on instruction-tuned models at 512 tokens, by a DeBERTa quality classifier and MMLU, with perplexity moving from 7.62 to 8.06 on Llama-3.1-8B and from 9.76 to 8.27 on Llama-3.2-1B (Table 7, p. 15) (Ardoyn, Schäfer, and Wunder 2026); we measured base models, continuation-only perplexity under the same model, our prompts, our probe training and four tuning keys. The negative result is also the probe's, not the watermark's: the exact test of Section 3 later attributed 97.5% of genuine texts at $\rho = 0.25$ (Section 5.3), where the perplexity ratio is 1.13. The attacks therefore run along the curve, $\rho \in \{0.25, 0.35, 0.50, 0.70\}$, rather than at one setting.

5.2 Forging the trained probe without the key

Study 1 asks whether a passive attacker with the open weights and $n \leq 1,024$ watermarked texts, and no detector access, can make the owner's probe accept new text, in three locked versions whose rules Table 1 lists (v0.4, with the repetition term on both sides, is the primary result). The probe's threshold held on the owner's null texts (pooled false-positive rate 1.4%, G1) but flagged 3.6% of the human continuations of the same prompts, and the watermark worked by the probe's own standard (genuine acceptance $\geq 20\%$, L1) only at $\rho \geq 0.35$ (8.0% at $\rho = 0.25$, which is descriptive in Study 1).

Route A recovers nothing. The activation-mean estimator (Section 2.3) never approached the key: the median cosine over the 8 keys stayed at or below 0.16 at every strength and budget up to $n = 1,024$, with the steered layer found for at most 3 of 8 keys at any budget (R1 fails at every level; Table A15; the full grid in Table A14). At $\rho = 0.35$ its v0.4 forgeries are **not practical** (upper bounds 11.5 and 10.0% against a bar of 14.3%); at 0.50 and 0.70 the verdict is **inconclusive**, because the per-key rates are bimodal: at $\rho = 0.70$ the median FA is 73.0% but its lower bound 8.5% (App. D).

Route B forges without the key. Footprint imitation at the attacker’s chosen layer was accepted by the probe at every working strength from $n = 64$ texts, at 42.5–100.0% against genuine rates of 32.0–100.0% (R2 practical at $n = 64$ at $\rho = 0.35, 0.50$ and 0.70), although its vector has a median cosine of at most 0.062 with the key (Table 3). Many of those texts were degenerate: steering at the first layers gave broken text the probe still accepted, and under v0.3’s perplexity-only condition most of Route B’s accepted texts at $\rho = 0.70$ were loops (exploratory readings; Table A15; App. E). With both conditions on both sides, the pre-registered v0.4 verdicts are: **Route B is practical at $\rho = 0.35$** (FA 19.5 [7.0, 35.0]% at $n = 256$ against a bar of 14.3%, its lower bound above the random key’s upper bound of 3.5%), and **inconclusive at 0.50 and 0.70**, where the probe accepts at least 99.0% of its texts but only 26–54% pass the fluency bars, so quality rather than detection limits the forgery (Figure 2). Generic steering alone never suffices (F3 no at every level). At $\rho = 0.35$ the watermark itself is weak by this detector (genuine texts accepted and fluent 28.5% of the time) and the bar is strict, those texts being close to unsteered text; at $\rho = 0.70$ the same relative bar is lenient, the genuine texts being themselves degraded, and one forgery that passed both bars read as word salad (exploratory; App. E).

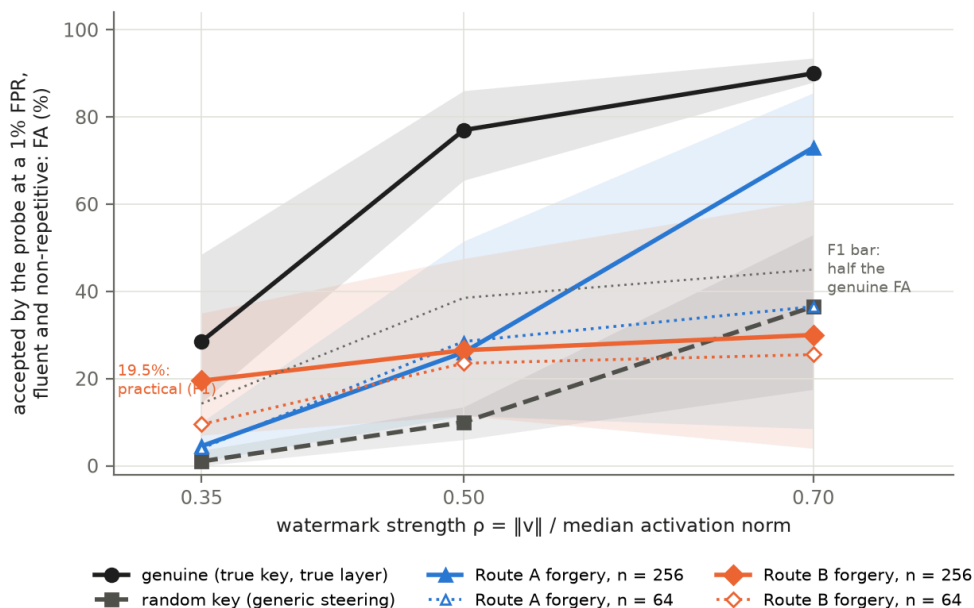


Figure 2. Fluent, non-repetitive acceptance by the owner’s probe at its 1% threshold against ρ for the genuine texts, the random-key control and the Route A and Route B forgeries at $n = 64$ and 256 (medians over 8 keys with 95% cluster-bootstrap bands; the dashed bar is half the genuine rate, rule F1).

Table 3. Forging the owner’s trained probe without the key (Study 1 v0.2 and v0.4; medians over 8 keys)

ρ	texts	v0.2 R2 verdict (plain acceptance)	v0.2 plain acceptance at $n = 64 / 256 / 1,024$ (%)	v0.4 FA at $n = 64 / 256$ (%) [95% CI]	F1 bar: half the oracle FA (%)	v0.4 F1 verdict	Route vector’s cosine with the key (v0.4, $n = 256$)
0.25	oracle (true key, true layer); descriptive, L1 fails	—	8.0 / — / —	—	—	watermark not working (L1)	—
0.35	oracle (true key, true layer)	—	32.0 / — / —	28.5 [14.5, 48.5]	—	ceiling	—
0.35	random key (generic steering)	—	1.5 / — / —	1.0 [0.0, 3.5]	—	F3 generic steering suffices: no	—
0.35	Route A: key recovery (activation mean)	inconclusive	11.0 / 3.5 / 4.5	4.0 [1.0, 11.5] / 4.5 [1.0, 10.0]	14.3	not practical	0.079

ρ	texts	v0.2 R2 verdict (plain acceptance)	v0.2 plain acceptance at $n = 64 / 256 / 1,024$ (%)	v0.4 FA at $n = 64 / 256$ (%) [95% CI]	F1 bar: half the oracle FA (%)	v0.4 F1 verdict	Route vector's cosine with the key (v0.4, $n = 256$)
0.35	Route B: footprint imitation	practical ($n=64$)	42.5 / 56.0 / 45.5	9.5 [1.5, 24.0] / 19.5 [7.0, 35.0]	14.3	practical ($n=256$)	0.047
0.5	oracle (true key, true layer)	—	86.0 / — / —	77.0 [65.5, 86.0]	—	ceiling	—
0.5	random key (generic steering)	—	13.0 / — / —	10.0 [6.0, 13.5]	—	F3 generic steering suffices: no	—
0.5	Route A: key recovery (activation mean)	not practical	23.0 / 20.0 / 20.5	28.5 [11.0, 52.5] / 26.0 [11.5, 51.5]	38.5	inconclusive	0.052
0.5	Route B: footprint imitation	practical ($n=64$)	99.0 / 99.5 / 100.0	23.5 [11.0, 46.0] / 26.5 [11.5, 47.5]	38.5	inconclusive	0.042
0.7	oracle (true key, true layer)	—	100.0 / — / —	90.0 [88.0, 93.5]	—	ceiling	—
0.7	random key (generic steering)	—	55.5 / — / —	36.5 [17.5, 53.0]	—	F3 generic steering suffices: no	—
0.7	Route A: key recovery (activation mean)	not practical	80.5 / 90.5 / 89.0	36.5 [7.5, 76.5] / 73.0 [8.5, 85.5]	45.0	inconclusive	0.000
0.7	Route B: footprint imitation	practical ($n=64$)	100.0 / 100.0 / 100.0	25.5 [1.0, 70.5] / 30.0 [4.0, 61.0]	45.0	inconclusive	0.062

Plain acceptance: the probe accepts the text at its 1% threshold (v0.2's rule). FA (v0.4's rule): accepted and no less fluent and no more repetitive than the key's own genuine watermarked texts (perplexity and seq-rep-4 at most their 95th percentiles). F1 practical: at some n the median FA reaches half the oracle's and its lower bound exceeds the random key's upper bound; not practical: both n 's upper bounds fall below the bar; otherwise inconclusive. v0.2's R2 uses plain acceptance with the same practical clause. The random-key row's verdict is F3.

Reading. We read these results as consistent with the probe detecting steering rather than the key (an inference): it accepts texts steered at other layers and in other directions, accepts degenerate text, and flags human text above its nominal rate. The scheme's authors note the related effect that "paraphrasing human text acts as a spoofing mechanism for our detector" (p. 5) ([Ardoyn, Schäfer, and Wunder 2026](#)), and the finding concerns the detector type as we re-implemented and re-trained it on Qwen2.5-1.5B at 256 tokens, not their deployed models. The claim that survives the pre-registered rules is narrow: on this model, at the one strength where quality is acceptable ($\rho = 0.35$), a passive attacker with 256 outputs and no key produces fluent, non-repetitive text that the probe accepts at more than half the genuine rate.

5.3 The exact test detects the key and rejects those forgeries

Study 3 re-scores Study 1's texts with the exact key test S4 (Section 3) on the same 8 keys and strengths, with the probe's numbers from Study 1 beside it on the same texts (Table 4; Figure 3). Calibration first: the pooled false-positive rate on 8 keys $\times$ 1,000 of the owner's unwatermarked texts was 0.81% (G1), every key was calibrated at 0.1–1.7% (G2), and the rate on human continuations was 0.68% (H1; highest key 1.6%), where the probe had flagged 3.6%. The test is exact on average over keys (Section 3); G2 is what makes it usable with one key, and on these keys it passed.

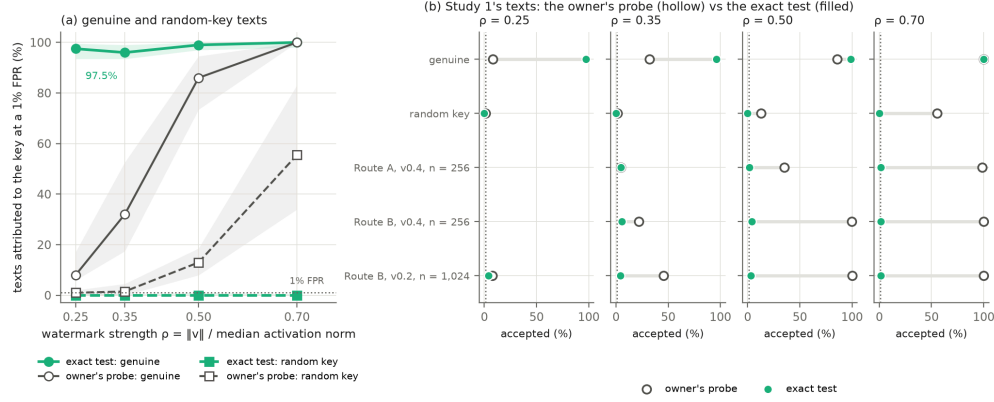


Figure 3. (a) Genuine and random-key texts attributed at an exact 1% false-positive rate by the exact test and by the probe, against ρ . (b) Study 1's texts accepted by the probe (hollow) and attributed by the exact test (filled), per forgery set and strength.

Table 4. The exact key test (S4) on the study keys: detection, random-key attribution and Study 1's forgeries re-scored (medians over 8 keys; 256-token texts; $p \leq 0.01$ with 999 null keys)

ρ	genuine texts attributed, exact test (%) [95% CI]	the probe on the same texts (%)	E1: exact – probe, points [95% CI]: verdict	random-key texts attributed: exact / probe (%)	genuine at $p \leq 0.001$ (%)	v0.4 Route A, n = 256: exact FA (%) [95% CI] (probe FA)	v0.4 Route B, n = 256: exact FA (%) [95% CI] (probe FA)	X1 bar: half the genuine exact FA (%)	X1 verdicts	X3 generic steering suffices
0.25	97.5 [93.5, 99.5]	8.0	+86.0 [+80.5, +90.0]: exact test more powerful	0.0 / 1.0	85.0	— (no v0.4 forgeries at 0.25)	—	44.0	—	no
0.35	96.0 [93.5, 99.0]	32.0	+62.0 [+46.0, +78.0]: exact test more powerful	0.0 / 1.5	93.0	4.5 [0.0, 14.0] (probe 4.5)	4.5 [1.5, 8.0] (probe 19.5)	43.2	A: not practical; B: not practical	no
0.5	99.0 [97.0, 100.0]	86.0	+13.5 [+3.0, +25.5]: exact test more powerful	0.0 / 13.0	98.5	1.5 [0.0, 32.5] (probe 26.0)	3.5 [0.0, 9.5] (probe 26.5)	45.0	A: not practical; B: not practical	no
0.7	100.0 [98.0, 100.0]	100.0	+0.0 [-2.0, +0.0]: no clear difference	0.0 / 55.5	98.0	1.0 [0.0, 22.0] (probe 73.0)	1.0 [0.0, 4.0] (probe 30.0)	45.0	A: not practical; B: not practical	no

Calibration: pooled false-positive rate 0.81% on 8 keys $\times$ 1,000 unwatermarked model texts (G1; range 0.3–2.5); every key calibrated at 0.1–1.7% (G2 bar 3%); 0.68% on 1,000 human continuations (H1; highest key 1.6%); at $p \leq 0.001$ with 9,999 null keys 0.11% on model text and 0.05% on human text. Exact FA: attributed by the exact test and fluent by v0.4's definition. X1 and X3 have the form of v0.4's F1 and F3 with the exact test in place of the probe. The probe's numbers are Study 1's, on the same texts.

Detection. The exact test attributed 97.5, 96.0, 99.0 and 100.0% of genuine 256-token texts at $\rho = 0.25$, 0.35, 0.50 and 0.70 (medians over 8 keys; P1 passes at every level), against the probe's 8.0, 32.0, 86.0 and 100.0% on the same texts; the paired differences are +86.0 [+80.5, +90.0], +62.0 [+46.0, +78.0] and +13.5 [+3.0, +25.5] points at the three lower strengths (E1: the exact test more powerful) and +0.0 [-2.0, +0.0] at $\rho = 0.70$ (no clear difference). Random-key texts, the generic-steering control, were attributed to the key 0.0% of the time at every strength (X3: no everywhere), where the probe accepted up to 55.5% of them at $\rho = 0.70$.

Forgeries. Scored by the exact test with v0.4’s fluency conditions, Study 1’s forgeries are **not practical for the median key at $\rho = 0.35$, 0.50 and 0.70, for both routes (X1)**: their exact fluent acceptance is 0.0–4.5% against bars of 43.2–45.0%, half the genuine exact FA. The one practical probe forgery, Route B at $\rho = 0.35$ and $n = 256$ with a probe FA of 19.5%, reaches 4.5% under the exact test; Route A at $\rho = 0.70$, at a probe FA of 73.0%, reaches 1.0%. Figure 3(b) shows the pattern over Study 1’s text sets: the probe’s acceptance of forgeries rises with ρ towards 100%, the exact test’s stays near the 1% line.

Per-key leakage (descriptive). The median hides where forgeries do pass. Over the $384 \text{ key} \times \text{strength} \times \text{forgery-set}$ cells, 35 have exact acceptance of 50% or more, and acceptance rises with the attacker’s recovered cosine (Spearman 0.68 for Route A and 0.56 for Route B over all cells; App. D). Route A’s forgeries pass mainly for the keys it partly recovered at high strength; some Route B forgeries pass at a cosine near zero, which we read as the copied footprint reproducing enough of that key’s effect on the output distribution for the gradient test to see (an inference). That steering directions differ in detectability and quality cost is noted by the scheme’s authors (p. 7) ([Ardoin, Schäfer, and Wunder 2026](#)); it is why per-key tables accompany every median (App. D).

Two consequences carry forward. Against these passive forgers with $n \leq 1,024$ texts, the exact test keeps the watermark’s value as provenance evidence for the median key, with more power at low strength, a false-positive rate checked per key and no acceptance of generic steering for the median key: a deployer should test the key, not the steering. And the protection is only as good as the key’s secrecy, since where an attack recovered much of the key its forgeries passed. Whether the key stays secret is Section 5.4.

5.4 The detector’s statistic steals the fixed key from 64 texts

Study 5’s gate P0 ran the attack that Section 3 predicts: Route A’ averages the exact test’s own standardised gradient over the attacker’s n observed texts, re-encoded through the open weights, and reads the key off the mean (Section 2.3; App. C). On the study keys it is **practical at $n = 64$ at $\rho = 0.35$ and 0.50** (Table 6; Figure 4). From 64 texts the median cosine with the key is 0.940 at $\rho = 0.35$ and 0.951 at 0.50 when the layer is given, and 0.932 and 0.920 when the attacker searches all 28 layers, which finds layer 14 for 6 of 8 keys at $\rho = 0.35$ and 6 of 8 at 0.50. Its fluent forgeries, made with the known-layer estimate, are then attributed as often as the owner’s genuine texts: FA 88.0 [76.5, 93.0]% against a genuine 86.5 [83.5, 90.5]% at $\rho = 0.35$, and 85.5 [82.5, 90.0]% against 90.0 [87.0, 92.5]% at 0.50, with the random-key control at 0.0 [0.0, 0.5]%; with the layer-search estimate, from 256 texts, they are attributed 86.0 [81.5, 90.5]% at $\rho = 0.35$ and 87.5 [80.5, 93.0]% at 0.50 (App. C.3). More texts change little (85.0% and 87.5% at $n = 1,024$; every key between 78 and 93% at 0.50). Study 1’s activation-mean estimator recovered nothing from the same 1,024 texts (median cosine ≤ 0.16 ; Section 5.2); the test’s statistic recovers the key from 64.

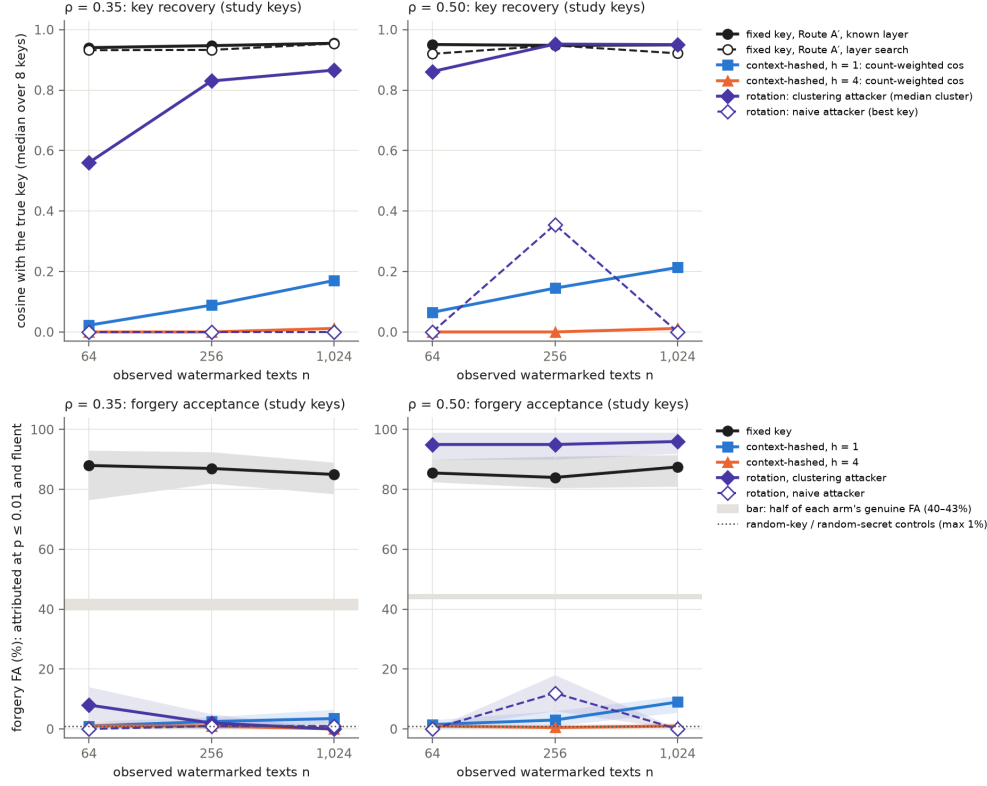


Figure 4. Key recovery (cosine with the true key) and forgery acceptance against the attacker’s budget n at $p = 0.35$ and 0.50 : the fixed key (Route A’ at the known layer and with the layer search), the context-hashed arms and rotation (the naive and clustering attackers), with the D1 bars and the random-key controls.

Table 6. The fixed key and the keyed arms under the gradient-averaging attacker: detection and stealing against the budget (Study 5; medians over 8 keys; rotation’s forgery sets have no per-key structure and are means)

arm and attacker	p	genuine detection, P1 (%)	key recovery at $n = 64 / 256 / 1,024$ (cosine with the key; count-weighted for the hashed arms)	forgery FA at $n = 64 / 256 / 1,024$ (%) [95% CI]	bar: half the genuine FA (%)	control FA (%) [95% CI]	verdict
fixed key ($h = 0$), Route A’	0.35	96.0	0.94 / 0.95 / 0.95	88.0 [76.5, 93.0] / 87.0 [82.0, 92.5] / 85.0 [78.5, 89.0]	43.2 (genuine FA 86.5)	0.0 [0.0, 0.5]	P0: practical ($n=64$)
context-hashed, $h = 1$	0.35	92.5	0.02 / 0.09 / 0.17	1.0 [0.0, 2.5] / 2.5 [0.0, 4.0] / 3.5 [2.0, 6.5]	42.0 (genuine FA 84.0)	1.0 [0.0, 1.5]	D1: stealing blocked at $n \leq 1024$
context-hashed, $h = 4$	0.35	90.0	0.00 / 0.00 / 0.01	1.0 [0.0, 2.5] / 1.0 [0.0, 2.0] / 0.0 [0.0, 0.0]	40.0 (genuine FA 80.0)	0.5 [0.0, 1.5]	D1: stealing blocked at $n \leq 1024$
rotation ($K = 8$), naive attacker	0.35	93.0 (union test; told the key: 96.0)	0.00 / 0.00 / 0.00 (best cos with any key)	0.0 [0.0, 0.0] / 1.0 [0.0, 3.0] / 1.0 [0.0, 3.0]	43.2 (the fixed key’s genuine FA)	0.0 [0.0, 0.5]	D1: stealing blocked at $n \leq 1024$
rotation ($K = 8$), clustering attacker	0.35	as above	4 of 8 keys (forging cluster cos 0.31) / 6 of 8 keys (forging cluster cos 0.00) / 6 of 8 keys (forging cluster cos 0.00)	8.0 [3.0, 14.0] / 2.0 [0.0, 5.0] / 0.0 [0.0, 0.0]	43.2	0.0 [0.0, 0.5]	D1: stealing blocked at $n \leq 1024$
fixed key ($h = 0$), Route A’	0.5	99.0	0.95 / 0.95 / 0.95	85.5 [82.5, 90.0] / 84.0 [80.5, 91.0] / 87.5 [81.0, 91.5]	45.0 (genuine FA 90.0)	0.0 [0.0, 0.5]	P0: practical ($n=64$)

arm and attacker	ρ	genuine detection, P1 (%)	key recovery at $n = 64 / 256 / 1,024$ (cosine with the key; count-weighted for the hashed arms)	forgery FA at $n = 64 / 256 / 1,024$ (%) [95% CI]	bar: half the genuine FA (%)	control FA (%) [95% CI]	verdict
context-hashed, $h = 1$	0.5	96.5	0.07 / 0.14 / 0.21	1.5 [0.0, 3.0] / 3.0 [1.0, 6.0] / 9.0 [5.5, 11.0]	44.0 (genuine FA 88.0)	1.0 [0.0, 1.0]	D1: stealing blocked at $n \leq 1024$
context-hashed, $h = 4$	0.5	95.5	0.00 / 0.00 / 0.01	1.0 [0.0, 2.0] / 0.5 [0.0, 1.5] / 1.0 [0.0, 2.0]	43.8 (genuine FA 87.5)	0.5 [0.0, 1.5]	D1: stealing blocked at $n \leq 1024$
rotation ($K = 8$), naive attacker	0.5	98.5 (union test; told the key: 99.0)	0.00 / 0.36 / 0.00 (best cos with any key)	0.0 [0.0, 0.0] / 12.0 [6.0, 18.0] / 0.0 [0.0, 0.0]	45.0 (the fixed key's genuine FA)	0.0 [0.0, 0.5]	D1: stealing blocked at $n \leq 1024$
rotation ($K = 8$), clustering attacker	0.5	as above	7 of 8 keys (forging cluster cos 0.95) / 8 of 8 keys (forging cluster cos 0.96) / 8 of 8 keys (forging cluster cos 0.96)	95.0 [90.0, 99.0] / 95.0 [90.0, 99.0] / 96.0 [92.0, 99.0]	45.0	0.0 [0.0, 0.5]	D1: practical ($n=64$)
post hoc, labelled (App. G): rotation, clustering attacker, forging with every cluster	0.35	as above	$n = 256$: 7 clusters with $\cos \geq 0.5$; the study's rule picked cos 0.00	$n = 256$: best cluster 96; largest cluster 88; the study's rule 1	43.2	—	7 of 7 recovered clusters reach the bar (not pre-registered; the D1 verdict stands)
post hoc, labelled (App. G): rotation, clustering attacker, forging with every cluster	0.35	as above	$n = 1,024$: 6 clusters with $\cos \geq 0.5$; the study's rule picked cos 0.00	$n = 1,024$: best cluster 90; largest cluster 89; the study's rule 1	43.2	—	5 of 6 recovered clusters reach the bar (not pre-registered; the D1 verdict stands)

FA: attributed at $p \leq 0.01$ (the keyed test for the hashed arms; the union test for rotation) and fluent by the arm's own genuine texts' bars. P0 and D1: practical at the first n where the FA reaches the bar with its lower bound above the control's upper bound; otherwise stealing blocked at $n \leq 1,024$. The control is a random key (fixed arm, rotation) or a random secret (hashed arms). The last two rows are the labelled post-hoc addendum (not pre-registered); the pre-registered rotation verdict at $\rho = 0.35$ stands as written.

This is the measured fact behind the paper's central inference (Section 3): the gradient the owner projects on the key to detect it is, averaged over outputs, an estimator of the key, so the detector's statistic is also the best estimator among those we tested (Section 2.3). Two bounds go with it. The budget comparison with token-level stealing carries an access caveat: our attacker needs 64 texts of 256 tokens, 16,384 tokens in all, where the token-level attacks use 30,000 responses, 2.2 million tokens or 2,000 texts (Section 2.2) (Jovanović, Staab, and Vechev 2024; Pang et al. 2024; Zhang et al. 2026), but those attackers query a black-box API, and ours re-encodes the outputs through the model that produced them, which only the open weights allow. And the strength matters: on the tuning keys the pilot that motivated Study 5 reached a median cosine of 0.80 from 4 texts and 0.95 from 16 at $\rho = 0.35$, but only 0.57 from 100 texts at $\rho = 0.70$ (a pilot, labelled; Table A5), where the study keys were not attacked; at that strength the attack is weaker and untested.

5.5 Scrubbing under the exact test

Study 2 scrubs the fixed key's genuine texts and scores the result with the exact test at matched quality (Section 4; Table 5; Figure 5): 100 genuine texts per key and strength, paraphrased by P-Qwen and P-Phi with the attacker's self-check and up to three attempts, or edited at 5% of the words by the true key (an insider bound), by the attacker's Route A estimate from Study 1, or at random. The test stayed calibrated on paraphrased null text (pooled false-positive rates 0.33% and 0.44%; every key at 0.0–0.6% and 0.0–1.4%) and attributed the unscrubbed originals at 96.0–100.0%.

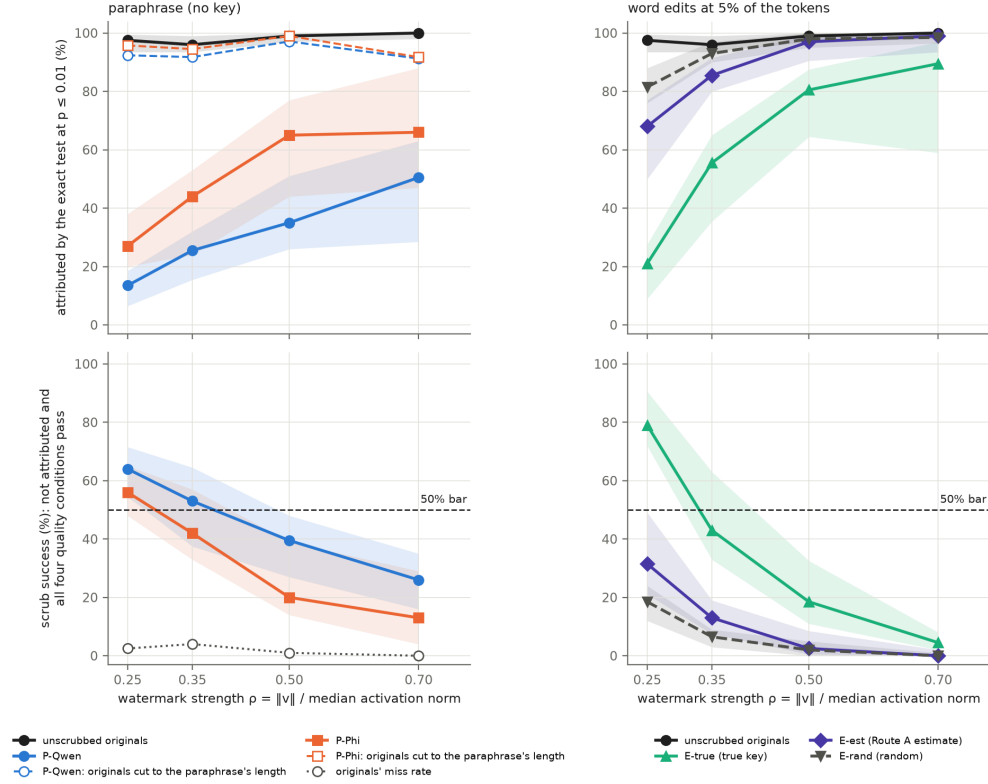


Figure 5. Detection after scrubbing (top) and scrub success (bottom) against ρ for the two paraphrasers, with the length-matched controls and the originals’ miss rate, and for word edits at 5% of the tokens with the true key, the attacker’s estimate and random positions; the dashed line is the 50% bar.

Table 5. Scrubbing the fixed key’s evidence under the exact test at matched quality (Study 2; medians over 8 keys; 100 genuine texts per key and strength)

ρ	method	detected after (%)	all four quality conditions pass (%)	scrub success (%) [95% CI]	keys $\geq$ 50%	pre-registered verdict (comparator)
0.25	unscrubbed originals	97.5	—	—	—	P1 pass
0.25	P-Qwen paraphrase	13.5	75.0	64.0 [54.5, 71.5]	7 of 8	S1 effective (originals’ miss 2.5 [0.5, 6.5])
0.25	P-Phi paraphrase	27.0	83.0	56.0 [48.0, 65.0]	7 of 8	S1 effective (originals’ miss 3.0 [0.0, 6.0])
0.25	E-true: 5% edits, true key	21.0	99.0	79.0 [72.0, 90.5]	8 of 8	S2 effective
0.25	E-est: 5% edits, Route A estimate	68.0	99.0	31.5 [21.0, 49.0]	2 of 8	S2 not effective
0.25	E-rand: 5% edits, random	81.5	99.5	18.5 [12.0, 24.0]	0 of 8	C1 E-est – E-rand: +14.5 [+5.0, +28.0]: the estimate helps
0.35	unscrubbed originals	96.0	—	—	—	P1 pass
0.35	P-Qwen paraphrase	25.5	74.0	53.0 [37.5, 64.5]	5 of 8	S1 effective (originals’ miss 4.0 [1.0, 6.5])
0.35	P-Phi paraphrase	44.0	79.0	42.0 [33.0, 57.0]	2 of 8	S1 inconclusive (originals’ miss 3.0 [0.0, 7.0])
0.35	E-true: 5% edits, true key	55.5	98.0	43.0 [33.0, 63.0]	3 of 8	S2 inconclusive
0.35	E-est: 5% edits, Route A estimate	85.5	98.5	13.0 [7.5, 19.0]	0 of 8	S2 not effective
0.35	E-rand: 5% edits, random	93.0	99.0	6.5 [3.0, 9.0]	0 of 8	C1 E-est – E-rand: +6.5 [+2.0, +11.0]: the estimate helps
0.5	unscrubbed originals	99.0	—	—	—	P1 pass

ρ	method	detected after (%)	all four quality conditions pass (%)	scrub success (%) [95% CI]	keys $\geq$ 50%	pre-registered verdict (comparator)
0.5	P-Qwen paraphrase	35.0	64.0	39.5 [27.0, 48.0]	1 of 8	S1 not effective (originals' miss 1.0 [0.0, 3.0])
0.5	P-Phi paraphrase	65.0	76.0	20.0 [14.0, 39.0]	1 of 8	S1 not effective (originals' miss 0.0 [0.0, 2.0])
0.5	E-true: 5% edits, true key	80.5	95.5	18.5 [11.0, 32.5]	1 of 8	S2 not effective
0.5	E-est: 5% edits, Route A estimate	97.0	93.5	2.5 [0.5, 8.5]	0 of 8	S2 not effective
0.5	E-rand: 5% edits, random	98.0	96.0	2.0 [0.0, 5.0]	0 of 8	C1 E-est – E-rand: +0.0 [-0.5, +7.0]: no clear difference
0.7	unscrubbed originals	100.0	—	—	—	P1 pass
0.7	P-Qwen paraphrase	50.5	49.5	26.0 [16.0, 35.0]	1 of 8	S1 not effective (originals' miss 0.0 [0.0, 2.0])
0.7	P-Phi paraphrase	66.0	52.0	13.0 [4.0, 29.0]	1 of 8	S1 not effective (originals' miss 0.0 [0.0, 2.0])
0.7	E-true: 5% edits, true key	89.5	41.5	4.5 [2.0, 8.0]	0 of 8	S2 not effective
0.7	E-est: 5% edits, Route A estimate	99.0	38.5	0.0 [0.0, 2.0]	0 of 8	S2 not effective
0.7	E-rand: 5% edits, random	98.5	39.0	0.0 [0.0, 1.0]	0 of 8	C1 E-est – E-rand: +0.0 [-0.5, +0.0]: no clear difference

Success: not attributed by the exact test at $p \leq 0.01$ and fluency, repetition, length and meaning within the owner's bars (§4). S1 and S2: effective if the median success is at least 50% and its lower bound exceeds the comparator's upper bound (the originals' miss rate for paraphrase; E-rand for the edits); not effective if the upper bound is below 50%; otherwise inconclusive. C1 is the paired difference E-est – E-rand over keys. Calibration after paraphrase: pooled false-positive rate 0.33% (P-Qwen) and 0.44% (P-Phi) on 8 keys $\times$ 1,000 paraphrased unwatermarked texts; every key calibrated (0.0–0.6% and 0.0–1.4%).

Paraphrase scrubs at the quality-acceptable strengths and not above them. At $\rho = 0.25$ both paraphraserers are **effective**: success 64.0 [54.5, 71.5]% for P-Qwen and 56.0 [48.0, 65.0]% for P-Phi against the originals' miss rates of 2.5 [0.5, 6.5]% and 3.0 [0.0, 6.0]%, with 7 and 7 of 8 keys at or above 50%. At $\rho = 0.35$ P-Qwen is **effective** (53.0 [37.5, 64.5]%; 5 of 8 keys) and P-Phi **inconclusive** (42.0 [33.0, 57.0]%). At $\rho = 0.50$ and 0.70 **neither is effective** (P-Qwen 39.5 and 26.0%; P-Phi 20.0 and 13.0%; one key in each cell reaches 50%). The evidence is removed, not shortened: the originals cut to the paraphrase's length are still attributed 91.2–99.0% of the time. At high strength quality is the other half of success: the share of P-Qwen's paraphrases passing all four conditions falls to 49.5% at $\rho = 0.70$.

The true key helps only where the watermark is weak. Edits of 5% of the words guided by the true key are **effective only** at $\rho = 0.25$ (success 79.0 [72.0, 90.5]%; all 8 keys), **inconclusive at 0.35** (43.0 [33.0, 63.0]%) and not effective above (at 10% of the words the key removes the evidence at a visible fluency cost; App. F). Edits guided by the attacker's Route A estimate are **never effective** (success 31.5% at $\rho = 0.25$, falling to 0.0%), although they beat random edits at $\rho = 0.25$ and 0.35 (C1: +14.5 [+5.0, +28.0] and +6.5 [+2.0, +11.0] points; the estimate helps) and not above.

Two secondary readings. The per-key false-positive rate disperses after paraphrase: over 1,000 fresh keys of the public distribution the mean stays at 1.02% and 0.94% (exact on average), but 7.8% and 6.5% of keys exceed 3% (maxima 22.6% and 19.5%), so a deployer should check their own key on paraphrased null text, the practical form of Section 3's "on average over keys" caveat. And the owner's trained probe accepts most of the same paraphrases at $\rho \geq 0.50$ where the exact test does not, which we record as a lead, not a result (App. F). The comparison with the authors' robustness figures is indirect: their detector, models, paraphraser and metric differ, and their Table 3 reports attribution accuracy between two steered

variants rather than detection against human text at a fixed false-positive rate (Ardoin, Schäfer, and Wunder 2026).

5.6 Keyed directions: context hashing and rotation, as transfers

Study 5 transfers two token-level defences to activation steering: keying the direction by the previous h tokens, the context hashing of Kirchenbauer et al. (Kirchenbauer et al. 2023, 2024), and rotating among K keys with a union test, the multiple-key design of the same authors (Kirchenbauer et al. 2023) as studied by Pang et al. and Gu et al. (Pang et al. 2024; Gu et al. 2024) and deployed by SynthID-Text (Dathathri et al. 2024); both were named as future work by the scheme’s authors (p. 7) (Ardoin, Schäfer, and Wunder 2026). The arms are $h = 1$, $h = 4$ and rotation among $K = 8$ keys, beside the fixed key, at $\rho = 0.35$ and 0.50 with the same 8 keys, against the attacker of Section 5.4 adapted to each arm (per observable context for the hashed arms; on the whole mixture, the naive attacker, or per cluster, the clustering attacker, for rotation; App. C) at $n \in \{64, 256, 1,024\}$, with Study 2’s paraphrase pipeline for the robustness cost. The gates passed (pooled false-positive rates 0.83, 0.95 and 1.00%; every key calibrated) and the keyed tests attributed 90.0–96.5% of their genuine texts, the union test 93.0 and 98.5% (P1; Table 6).

Stealing (D1). Both hashed arms **block this attacker at $n \leq 1,024$** : the per-context attacker’s forgeries are attributed at most 9.0% (medians; upper bounds at most 11.0%) against bars of 40–44%, half the arms’ genuine FA (Table 6; Figure 4); its per-context estimates stay poor (count-weighted cosine 0.213 for $h = 1$ at $n = 1,024$, $\rho = 0.50$; App. C). The verdict is bounded by the budget, not by the scheme: $h = 1$ ’s acceptance rises with n at $\rho = 0.50$ ($1.5 \rightarrow 3.0 \rightarrow 9.0\%$), and the token-level literature steals context-hashed schemes with tens of thousands of responses (Section 2.2). Rotation blocks the naive attacker at both strengths but **falls to the clustering attacker at $\rho = 0.50$, practical at $n = 64$** : it separates the mixture into the keys (7 of 8 recovered at cosine ≥ 0.5 from 64 texts, 8 of 8 from 256) and the union test attributes 95–96% of its forgeries. At $\rho = 0.35$ the pre-registered verdict is **stealing blocked at $n \leq 1,024$** , and it stands as written; its reason is the attacker’s pre-set rule for choosing the forging cluster, which at $n = 256$ and $1,024$ picked an outlier cluster with cosine 0.000 to every key while the other clusters recovered 6 of 8. A post-hoc addendum, labelled and not pre-registered (App. G; the last rows of Table 6; the hollow marker in Figure 6), forged with every cluster in turn: the best cluster’s forgeries are attributed 96% at $n = 256$ and 90% at $n = 1,024$, and 5 of the 6 recovered clusters reach the bar. Read with the pre-registered result, rotation among 8 keys multiplies this attacker’s cost by about K against averaging and by nothing against clustering, at both strengths.

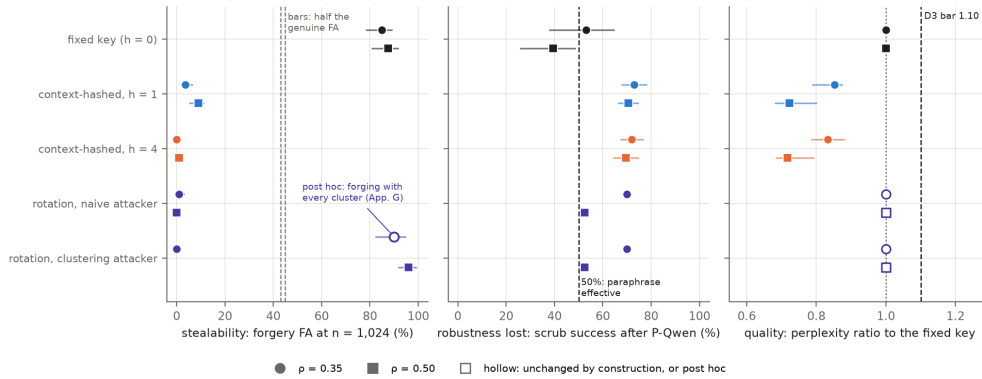


Figure 6. The frontier per arm at $\rho = 0.35$ and 0.50 : forgery acceptance at $n = 1,024$ (stealability), scrub success after P-Qwen (robustness lost) and the perplexity ratio to the fixed key (quality), with the materiality bars. Rotation’s clustering point at $\rho = 0.35$ is shown twice: the pre-registered rule’s value and the addendum’s every-cluster value (hollow, post hoc).

Robustness (D2). The hashed arms lose most of the paraphrase robustness the fixed key has (Table A16; Figure 6). After P-Qwen with the attacker’s self-check, scrub success is 73.0 and 70.5% against $h = 1$ and 72.0 and 69.5% against $h = 4$, where the fixed key on the same prompts and keys was scrubbed 53.0 and 39.5% of the time (Section 5.5). The paired differences are **material** for $h = 1$ at both strengths (+22.0 [+6.0, +36.5] and +32.0 [+21.0, +44.0] points) and for $h = 4$ at $\rho = 0.50$ (+32.5 [+19.0, +45.5]), and **inconclusive** for $h = 4$ at $\rho = 0.35$ (+18.0 [+7.5, +33.5]: the median falls under the 20-point bar although the interval excludes zero). Attribution after paraphrase falls to 1.0–4.0% for the keyed arms against the fixed key’s 25.5 and 35.0%: a paraphrase changes most contexts, so the content shift that survives it no longer lines up with any per-context key the owner can test (an inference; the same mechanism denies the attacker a stable direction to average). Rotation keeps the fixed key’s robustness: the union test’s loss on Study 2’s paraphrases is +15.0 [+9.5, +18.5] and +13.0 [+8.5, +17.5] points, **immaterial** by the 20-point bar.

Quality (D3). Context keying costs nothing by our proxy and improves it, with perplexity ratios to the fixed key’s texts on the same prompts of 0.72–0.85 (medians; no key above the 1.10 bar; **immaterial** everywhere), which is an improvement by that measure and not a quality claim (Section 4); rotation’s quality is unchanged by construction.

The point of difference with the token-level evidence. That evidence on multiple keys is mixed (Section 6); our rotation result sits with Jovanović et al.’s ([Jovanović, Staab, and Vechev 2024](#)) under this access model rather than with Pang et al.’s ([Pang et al. 2024](#)), whose removal attack is untested here, so the no-free-lunch trade-off transfers only in this form: the context-width trade-off between stealing and paraphrase robustness reappears, rotation buys a constant factor against averaging and nothing against clustering, and immunity is claimed for no arm.

5.7 The trade-off, read across the studies

Read together, the studies describe one curve for the fixed key on this model (an interpretation, labelled; Figure 6). The strengths at which the exact test detects the key and paraphrase fails to scrub it, $\rho \geq 0.50$, cost perplexity (ratios 1.62 and 2.29 against unsteered text; Section 5.1); the strengths with acceptable perplexity, $\rho \leq 0.35$, are scrubbed by an off-the-shelf paraphrase (Section 5.5). The published probe fails the detectability criterion at the low strengths and is forgeable without the key at 0.35 (Section 5.2); the exact test rescues detection at every strength (Section 5.3), but its own statistic steals the fixed key from 64 texts (Section 5.4). Context keying moves the stealing corner at the price of the robustness corner; rotation keeps robustness and quality and falls to clustering (Section 5.6). That watermark designs trade robustness, quality and security against one another is the token-level literature’s main theme ([Pang et al. 2024](#); [Shen, Huang, and Wan 2025](#); [Jovanović, Staab, and Vechev 2024](#)) and is noted by the scheme’s authors (p. 2) ([Ardoyn, Schäfer, and Wunder 2026](#)); the contribution here is the measured curve along ρ for this family, on one model and against one attacker family.

6 Related work

Stealing, forgery and scrubbing of token-level watermarks. Jovanović et al. ([Jovanović, Staab, and Vechev 2024](#)) define stealing as “the act of reverse-engineering a watermark by querying the API of the watermarked LLM” (p. 1), with a black-box attacker (p. 3) and “ $n = 30,000$ responses of token length ≤ 800 ” (p. 6), spoof and scrub KGW-style schemes from the same estimate, and find that “the supposed tradeoff between spoofing and scrubbing robustness ... in fact does not hold” (p. 9). Pang et al. ([Pang et al. 2024](#)) state the trade-offs as guidelines, “Robust watermarks are inherently vulnerable to spoofing attacks and are not suitable as proof of content authenticity alone” (p. 5) and more keys defend against stealing at a cost in removal robustness (p. 7), stealing from “2.2 million tokens in total” (p. 6). Gu et al.

([Gu et al. 2024](#)) learn a watermark by distillation and conclude that “watermarking should not be used to attribute provenance or blame to a specific LM” (p. 2). Zhang et al. ([Zhang et al. 2026](#)) steal with “2,000 victim watermarked texts” (p. 8); MarkSec ([Li et al. 2026](#)) fixes the attack definitions we adopt (p. 3) and excludes model-parameter schemes by scope (p. 2); SEEK ([Shen, Huang, and Wan 2025](#)) calls the scrubbing–forgery trade-off “widely acknowledged” (p. 1) and moves its boundary at the token level; paraphrase as an attack on detectors is Krishna et al. ([Krishna et al. 2023](#)). What remains: every attack in the evaluations cited is token-level and black-box, none touches an activation scheme, and our attacker’s white-box re-encoding changes the budget in kind (Section 2.2); the provenance conclusion of Gu et al. and Pang et al. is drawn here for this family (Section 5.3), citing them.

Context hashing, key rotation and their trade-offs. Both transferred mechanisms originate with Kirchenbauer et al. ([Kirchenbauer et al. 2023](#)): green lists from “a pseudorandom number generator seeded with previous tokens” (p. 4), a leak at $h = 1$ that brute force can exploit (p. 6), the robustness cost of large h , “attack amplification” (p. 6), and the multiple-key design, “have $k > 1$ different hidden keys, and to randomly choose one for each generation. At detection time, we run k tests, one for each of the possible keys” (p. 7), defended by the absence of bias when averaging over many generations (p. 7). Our rotation arm is this design transferred; our clustering attacker is the counter to that averaging argument. Kirchenbauer et al. ([Kirchenbauer et al. 2024](#)) find that “a small context width h provides the best robustness to machine paraphrasing” (p. 21) while secrecy wants a large h or “sufficiently many keys” (p. 17), the trade-off our $h = 1$ and $h = 4$ arms reproduce at the activation level. SynthID-Text deploys both mechanisms ([Dathathri et al. 2024](#); [Hugging Face 2026](#)) and names stealing, forgery and scrubbing as “an area of ongoing research” (p. 6). On multiple keys the token-level evidence is mixed: three keys cut Pang et al.’s stealing success from 91% to 13% (p. 6), while Jovanović et al. kept forgery success of 0.68–0.81 with $k = 2$ –4 keys switched per response (Table 14, p. 20), “not a viable defense against our attacker” (p. 20); Gu et al. note the multiple-testing cost (pp. 19–20), and Zhang et al. name random key selection as a defence with a detection cost (p. 18).

Model-internal watermarks and keyed directions. Self-Recognition ([Ardoin, Schäfer, and Wunder 2026](#)) is the scheme analysed (Section 2.1); its security rests, by its authors’ statement, on the secrecy of the vector and layer and is left to future work (p. 7), and they report that “paraphrasing human text acts as a spoofing mechanism for our detector” (p. 5) and that “Some vectors induce more severe degradations in generation quality, while others are more easily detectable” (p. 7), which our per-key tables bear out. SLAM ([Harel-Canada and Sahai 2026](#)) steers public sparse-autoencoder features chosen per document by a keyed hash (p. 5), reads a document identifier at detection, reports that “DIPPER reduces TPR to 12%/10% on 2B/9B” (p. 7) and targets “provenance attribution rather than tamper-proof cryptographic authentication” (p. 2); the words stealing, spoofing, forgery and threat model do not occur in it. AWM ([Aremu et al. 2026](#)) fine-tunes a keyed Gaussian direction per layer to monitor harmful outputs and finds that “attacks optimized against one surrogate key do not reliably transfer against other (independent) keys” (p. 2) for an adaptive evader who “cannot query the detector” (p. 3): the nearest keyed-direction design, under a different threat. GaussMark ([Block, Rakhlin, and Sekhari 2025](#)) perturbs weights, needs them for detection, “a significantly stronger access model” (p. 19), and leaves its per-token-key variant untested (p. 48). What remains: no member of the family has a measured key-recovery or forgery result under an attacker who holds the weights, and the paraphrase and evasion figures that SLAM’s and AWM’s authors report are robustness measurements of their own schemes, not scrubbing evaluations at matched quality under a per-key-calibrated test; SLAM and AWM are untested here.

Score tests and randomisation tests. Our detector is GaussMark’s score statistic ([Block, Rakhlin, and Sekhari 2025](#); [Rao 1948](#)) for an additive activation vector, with Monte Carlo p-values ([Phipson and Smyth 2010](#)) from resampled sparse keys in place of the Gaussian null (Section 3). GaussMark’s guaran-

tees “are valid for any fixed $x \in \mathcal{X}$, but are marginalized over the key ξ ” (p. 11); the per-key calibration gate we add is a practice, not a theorem, and the dispersion of per-key false-positive rates after paraphrase (Section 5.5) is its reason. Concurrently and outside watermarking, Chen et al. (Chen et al. 2026) recover a known residual-stream steering vector from a steered model’s outputs by distillation and report the first-order relation of Section 3 in their setting, the initial gradient “point[s] toward $F\Delta_T$ rather than Δ_T ” (p. 1), which iterative optimisation then inverts; our Route A’ (Section 5.4) is that first gradient, standardised and averaged over the observed outputs, with no inversion.

Quality bars. Self-Recognition’s appendix observes that “a highly repetitive, degenerate loop often yields low PPL” (p. 14); SLAM scores quality on four axes (p. 8); MarkSec relies on a judge model with “no independent human annotation” (p. 8). Our conditions pair perplexity with a repetition term on both the attacker’s and the evaluator’s side, add length and meaning conditions for scrubbing, and read fixed samples (Section 2.2, Section 4, App. E); they remain proxies, and no human judgement was collected.

7 Limitations and future work

What was measured, and on what. Every claim is bounded to one scheme as re-implemented by us (Self-Recognition’s construction, not its released weights or deployed detector), one base model (Qwen/Qwen2.5-1.5B, layer 14 of 28), 8 study keys drawn from the scheme’s own sparse key distribution, 256-token continuations of news-like prompts, and the strengths $\rho \in \{0.25, 0.35, 0.50, 0.70\}$ (Study 5: 0.35 and 0.50). Rates are medians over keys with cluster-bootstrap intervals, except rotation’s forgery rates, which are means over one corpus (Section 5.6), and the per-key tables of Appendix D show that keys differ: one key is detected far less often than the others at the highest strength, a few keys leak through the exact test where most do not, and the keyed arms’ robustness cost varies by tens of points across keys. Where a pre-registered rule did not reach a verdict we say so: fluent forgery of the probe at $\rho = 0.50$ and 0.70 , P-Phi paraphrase and true-key edits at 0.35 , and the robustness cost of $h = 4$ at 0.35 are inconclusive, not negative.

The attackers and their budgets. We tested three passive attacker families (Routes A, A’ and B; the per-context and clustering attackers of Section 5.6), each holding the open weights and at most $n = 1,024$ watermarked texts and none querying the detector. “Stealing blocked at $n \leq 1,024$ ” therefore means blocked for these attackers at this budget, not immunity: the $h = 1$ arm’s forgery acceptance rises with n ($1.5 \rightarrow 3.0 \rightarrow 9.0\%$ at $\rho = 0.50$), and the token-level literature steals context-hashed schemes with far larger budgets under a different access model (Section 5.4). We did not test an attacker who knows the exact test and optimises against it, one who combines paraphrase with key-guided edits, one who queries the detector, the repeated-query removal attack that Pang et al. (Pang et al. 2024) run against multiple keys, or human editing. At $\rho = 0.70$ the key-recovery attack was weaker on tuning keys and was not run on the study keys.

Quality is a proxy. Fluency and faithfulness are measured by perplexity under the unsteered model, a repetition term, length and embedding similarity, with samples read under pre-set rules (Appendix H); no human judgement was collected, and the scheme’s own quality classifier and MMLU evaluation were not computed. Because the relative bar is lenient at the strongest watermark, where the genuine text is itself degraded (Appendix E), fluent acceptance at $\rho = 0.70$ should not be read as high-quality forgery.

The reproduction and the exactness statement. The published strength did not reproduce *under our conditions*: base rather than instruction-tuned models, continuation-only perplexity under the same model, our prompts and our probe training. The authors’ numbers are beside ours in Section 5.1, and we draw no conclusion about their claims beyond those conditions. The exact test’s level is exact on average over keys; the per-key gate held for these 8 keys on unparaphrased text, but after paraphrase 7.8–6.5% of

freshly drawn keys exceeded a 3% false-positive rate (Section 5.5, Appendix D), which is why a deployer should check the one key it holds.

Novelty is at the search level. “To our knowledge” means that no attack on Self-Recognition, SLAM or AWM appeared in the citing papers, listings and repositories we searched through October 2026; search absence is not proof of novelty. SLAM-style and AWM-style designs were discussed, not tested.

Future work. Five studies were specified but not run and would complete the picture: a SLAM-style keyed feature-steering target under the same protocols (the planned Study 4); the $h = 1$ acceptance curve beyond $n = 1,024$ texts, where the token-level budgets lie; the per-position form of the keyed test with $h = 0$ on the study keys (measured so far on tuning keys only; Appendix G); Pang et al.’s removal attack against rotation; and the 8B scale at which the scheme’s authors report their results. An adaptive attacker against the exact test, and a human or model-judged coherence rating in place of the perplexity proxies, are the two measurements whose absence most limits the conclusions above.

8 Reproducibility, ethics and disclosure

Pre-registration and locks. The 6 studies were run from protocols fixed and hashed before any outcome existed (locked between 2026-09-26 and 2026-10-02), together with every code file the run imports and every reused input; the runners re-hash these files and refuse to run on a mismatch. A correction became a new version with the earlier results kept (Study 1 v0.2, v0.3 and v0.4 are all reported), the pilots that fixed the designs were run on tuning keys that no study used, and the one post-hoc analysis is labelled wherever it appears (Section 5.6, Appendix G). Table 1 gives each study’s lock date, hash prefix, commit and run time; Appendix A gives the full table and one paragraph per protocol.

Code and data availability. The protocols, lock files, validation reports, run manifests, result files and every script that generated a number, table or figure in this paper are released in a public repository (<https://github.com/SHENSHENZYC/activation-watermark-security>; code under the MIT licence, text and figures under CC BY 4.0), without model weights or the generated corpora. Prompts come from the C4 realnewslake validation shard ([Raffel et al. 2020](#)) (ODC-BY 1.0, under the Common Crawl terms): 8,580 usable units, of which 6,896 were available for development and 1,684 were held out behind a loader that refuses to open them, and were never used. A WikiText-103 set ([Merity et al. 2017](#)) was frozen as a second domain and is used by none of the six studies. No article text is redistributed; the repository holds identifiers and hashes only. Models were pinned to a revision and hashed (Appendix A): Qwen2.5-1.5B (apache-2.0) ([Qwen Team 2024a](#), [2024b](#)), the paraphraser Qwen2.5-1.5B-Instruct (Apache-2.0) ([Qwen Team 2024c](#)) and Phi-3.5-mini-instruct (MIT) ([Abdin et al. 2024](#)), Llama-3.2-1B and 3B (Llama 3.2 Community Licence) ([Grattafiori et al. 2024](#)) for calibration only, and all-mpnet-base-v2 (Apache-2.0) ([Reimers and Gurevych 2019](#)) for embedding similarity.

Compute. Everything ran on one Apple-silicon laptop (M5, 16 GB unified memory; PyTorch 2.14.0 on MPS). The six locked runs took 91.1 hours in total (22.4, 6.4, 6.5, 2.9, 26.6 and 26.3 h), plus the reproduction pilot, three calibrations and three design pilots; no cloud compute was used.

Ethics. The scheme analysed is a published research method, re-implemented from its paper and public code on open-weight models; no deployed system was attacked, no detector was queried, and no private data was used. The attacks measured are the removal and forgery attempts that evaluation guidance names (Appendix I), and the forgeries exist only as research artefacts: the generated corpora are not released, and the only generated texts in the repository are the samples quoted in Appendix H and in the study reports. Nothing here assesses any provider’s deployment or compliance.

Use of AI tools. The text and code were drafted with an AI system under the author’s direction, as stated in the acknowledgements after Section 9.

9 Conclusion

A deployer of an activation-steering watermark of this family learns four things from these studies, each bounded to one model, one layer, 8 keys, 256-token texts and a passive attacker with the open weights and at most 1,024 outputs. The published detector type, a trained probe, is forgeable without the key at the only strength where quality is acceptable, and, on our reading, it detects steering rather than the key (Section 5.2); testing the key directly with an exact, per-key-calibrated score test rescues detection at every strength and blocks those forgeries. That same statistic is the best estimator among those we tested, so the fixed key is recovered from 64 outputs at the known layer, or from 256 with a layer search, and forged, and an off-the-shelf paraphrase removes the evidence at the strengths with acceptable perplexity. Keying the direction by context, a transfer from token-level watermarking, stops that attacker at the budget tested at the price of most of the paraphrase robustness; rotating among keys keeps robustness and quality but buys only a constant factor against a clustering attacker (pre-registered at $\rho = 0.50$, post hoc at 0.35). If such watermarks are to serve as the robust, reliable and secure provenance mechanisms that regulators now describe, they will need keyed verification tools whose false-positive rate is checked per key, and whose security is measured against the attacker who holds the weights.

Acknowledgements and use of AI tools

This work received no external funding. The text and code of this paper were drafted by Claude (Anthropic) under the author’s direction and under the pre-registered protocols described in Section 4 and Appendix A; the author designed the studies, took every decision recorded in the project’s decision log, revised every section, verified every number against the locked outputs and takes full responsibility for the content. Codex (OpenAI) assisted with the final review and revision; the author reviewed and approved the changes. No AI system is an author. The language models that are part of the method (the watermarked model and the two paraphrasers) are described in Section 4.